Dual Matching Coefficients for the Continuous Hahn Weight $W(x) = 2\pi^2 \operatorname{sech}(2\pi x)$

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1 The Weight and Its Polynomials

The continuous Hahn weight with parameters $a = c = \frac{1}{4}$, $b = d = \frac{3}{4}$ is

$$W(x) = \Gamma(\tfrac{1}{4} + ix)\,\Gamma(\tfrac{3}{4} + ix)\,\Gamma(\tfrac{1}{4} - ix)\,\Gamma(\tfrac{3}{4} - ix) = \left|\Gamma(\tfrac{1}{4} + ix)\,\Gamma(\tfrac{3}{4} + ix)\right|^2.$$

By the reflection formula, with $s = \frac{1}{4} + ix$:

$$\Gamma(\tfrac{1}{4} + ix)\,\Gamma(\tfrac{3}{4} - ix) = \frac{\pi}{\sin\pi(\frac{1}{4} + ix)},$$
$$\Gamma(\tfrac{3}{4} + ix)\,\Gamma(\tfrac{1}{4} - ix) = \frac{\pi}{\sin\pi(\frac{3}{4} + ix)},$$

and since $\sin(\frac{\pi}{4} + i\pi x)\,\sin(\frac{3\pi}{4} + i\pi x) = \frac{1}{2}\sin(\frac{\pi}{2} + 2i\pi x) = \frac{1}{2}\cosh(2\pi x)$, we obtain the elementary form

$$W(x) = \frac{2\pi^2}{\cosh(2\pi x)}. \tag{1}$$

This is the *hyperbolic-secant weight*, an even, rapidly decaying function with poles at $x = \pm\frac{i}{4} + \frac{in}{2}$ ($n \in \mathbb{Z}$).

1.1 Moments and Euler Numbers

Since W is even, all odd moments vanish. The even moments are

$$\mu_{2n} = \int_{\mathbb{R}} x^{2n}\,W(x)\,dx = \frac{\pi^2\,|E_{2n}|}{16^n}, \tag{2}$$

where E_{2n} are the Euler numbers ($E_0 = 1$, $E_2 = -1$, $E_4 = 5$, $E_6 = -61$, $E_8 = 1385$, …; OEIS A000364).

Derivation. Substituting $u = 2\pi x$ gives $\mu_{2n} = 2\pi^2(2\pi)^{-(2n+1)}\int u^{2n}\operatorname{sech} u\,du$. The known identity $\int_{\mathbb{R}} u^{2n}\operatorname{sech} u\,du = \pi\,|E_{2n}|\,(\pi/2)^{2n}$ (which follows from the Fourier transform of sech being $\pi\operatorname{sech}(\pi\xi/2)$ and comparing Taylor coefficients) yields (2) after simplification. □

1.2 Monic Orthogonal Polynomials

The monic orthogonal polynomials p_n for W satisfy

$$p_{n+1}(x) = x\,p_n(x) - \frac{n^2}{16}\,p_{n-1}(x), \qquad p_0 = 1, \quad p_1 = x. \tag{3}$$

The recurrence coefficients $\alpha_n = 0$ (by evenness) and $\beta_n = n^2/16$ were computed exactly from the Euler-number moments via the Chebyshev algorithm with rational arithmetic, and verified to $n = 11$.

Remark 1.1. *These are rescaled Meixner–Pollaczek polynomials. The standard monic MP($\lambda = \frac{1}{2}$, $\phi = \frac{\pi}{2}$) polynomials $q_n(y)$ satisfy $q_{n+1} = y\,q_n - (n^2/4)\,q_{n-1}$ and are orthogonal for* $\operatorname{sech}(\pi y)$. *Scaling $y = 2x$ and $p_n(x) = 2^{-n} q_n(2x)$ converts* $\operatorname{sech}(\pi y)$ *to* $\operatorname{sech}(2\pi x)$ *and $\beta_n^{MP} = n^2/4$ to $\beta_n = n^2/16$.*

The norms (with respect to W/π^2) are $h_n/\pi^2 = \prod_{j=1}^{n}(j^2/16) = (n!)^2/16^n$, so the orthonormal polynomials are $\widehat{p}_n = p_n/\sqrt{h_n} = p_n \cdot 4^n/(\pi\, n!)$.

2 The Dual Split and Matching Matrices

The weight factors analytically:

$$W(x) = w_+(x)\, w_-(x), \qquad w_+(x) = \Gamma(\tfrac{1}{4} + ix)\,\Gamma(\tfrac{3}{4} + ix), \qquad w_-(x) = \overline{w_+(x)}.$$

This defines dual functions $F_m = w_+\,\widehat{p}_m$ and $G_m = w_-\,\widehat{p}_m$ with $\langle F_m, G_k\rangle = \delta_{mk}$.

We study two matching matrices: the *orthogonal-function matching*

$$b_{mk} = \int_{\mathbb{R}} \widehat{p}_m(x)\,\widehat{p}_k(x)\; W(x)^{3/2}\, dx,$$

with kernel $W^{3/2} = (2\pi^2)^{3/2}\operatorname{sech}(2\pi x)^{3/2}$ (which has branch points at $x = \pm i/4$), and the *dual-split matching*

$$c_{mk} = \int_{\mathbb{R}} \widehat{p}_m(x)\,\widehat{p}_k(x)\; w_+(x)\, W(x)\, dx,$$

whose kernel is $\Gamma(\frac{1}{4}+ix)^2\,\Gamma(\frac{3}{4}+ix)^2\,\Gamma(\frac{1}{4}-ix)\,\Gamma(\frac{3}{4}-ix)$ — a six-Gamma integral (three pairs).

By the symmetry $x \to -x$, both kernels enjoy a *checkerboard* structure: $b_{mk} = 0$ when $m+k$ is odd (verified to 10^{-58}), while c_{mk} is real when $m+k$ is even and purely imaginary when $m+k$ is odd (worst violation $< 3 \times 10^{-58}$).

3 Barnes Evaluation of c_{00}

Theorem 3.1. *The base entry of the dual-split matching matrix is*

$$c_{00} = 4\sqrt{\pi}\,U(1,1,2) = 4\sqrt{\pi}\,e^2\,E_1(2) = 2.56175319378357597\ldots, \tag{4}$$

where U is Tricomi's confluent hypergeometric function and $E_1(z) = \int_z^\infty e^{-t}/t\,dt$ is the exponential integral.

Proof. Since $\widehat{p}_0 = 1/\pi$, we have $c_{00} = (1/\pi^2)\int w_+\, W\, dx$. Legendre's duplication formula gives

$$\Gamma(\tfrac{1}{4}+ix)\,\Gamma(\tfrac{3}{4}+ix) = \sqrt{\pi}\;2^{1/2-2ix}\,\Gamma(\tfrac{1}{2}+2ix), \tag{5}$$

$$\Gamma(\tfrac{1}{4}-ix)\,\Gamma(\tfrac{3}{4}-ix) = \sqrt{\pi}\;2^{1/2+2ix}\,\Gamma(\tfrac{1}{2}-2ix). \tag{6}$$

Substituting into the six-Gamma integrand:

$$w_+(x)\,W(x) = \left[\sqrt{\pi}\;2^{1/2-2ix}\,\Gamma(\tfrac{1}{2}{+}2ix)\right]^2 \cdot \sqrt{\pi}\;2^{1/2+2ix}\,\Gamma(\tfrac{1}{2}{-}2ix),$$

which simplifies to

$$w_+\,W = \pi^{3/2}2^{3/2}\cdot 2^{-2ix}\,\Gamma(\tfrac{1}{2}{+}2ix)^2\,\Gamma(\tfrac{1}{2}{-}2ix).$$

The substitution $u = 2x$ then gives

$$\int_{\mathbb{R}} w_+ W\, dx = \pi^{3/2}\sqrt{2} \int_{\mathbb{R}} \Gamma(\tfrac{1}{2} + iu)^2\, \Gamma(\tfrac{1}{2} - iu)\, e^{-iu \ln 2}\, du.$$

The shifted three-Gamma integral (Proposition 2.1 of [1]) with $a = b = c = \frac{1}{2}$ and $s = -\ln 2$ (so $e^{-s} = 2$ and $e^{-as} = \sqrt{2}$) gives

$$\int_{\mathbb{R}} \Gamma(\tfrac{1}{2} + iu)^2\, \Gamma(\tfrac{1}{2} - iu)\, e^{-iu \ln 2}\, du = 2\pi\sqrt{2}\, U(1, 1, 2).$$

Hence $\int w_+ W\, dx = 4\pi^{5/2}\, U(1, 1, 2)$, and $c_{00} = 4\sqrt{\pi}\, U(1, 1, 2)$. □

The identity $U(1, 1, z) = e^z E_1(z)$ follows from the integral representation $U(1, 1, z) = \int_0^\infty e^{-zt}/(1 + t)\, dt = e^z \int_z^\infty e^{-t}/t\, dt$.

4 The Matching Coefficients

4.1 Matrix (a): $\operatorname{sech}^{3/2}$ kernel

Table 1 shows the 6×6 upper-left block of b_{mk}. The vanishing of entries with $m+k$ odd confirms the even-weight parity selection rule.

Table 1: Orthogonal-function matching coefficients b_{mk} ($m + k$ odd entries are zero).

b_{mk}	$k = 0$	1	2	3	4	5
$m = 0$	3.3889	0	−0.8216	0	0.4223	0
1		1.7456	0	−0.7759	0	0.4628
2			0.9926	0	−0.6059	0
3				0.7020	0	−0.4996
4					0.5319	0
5						0.4359

4.2 Matrix (b): Six-Gamma (dual-split) kernel

Table 2 shows the 6×6 upper-left block of c_{mk}. The checkerboard pattern holds: entries with $m + k$ even are real, entries with $m + k$ odd are purely imaginary.

Table 2: Dual-split matching coefficients c_{mk} ($m + k$ even: real; $m + k$ odd: displayed as coefficient of i).

c_{mk}	$k = 0$	1	2	3	4	5
$m = 0$	2.5618	−1.3709	−1.0109	0.7799	0.6420	−0.5428
1		0.5399	−0.4020	−0.4647	0.4056	0.3824
2			0.3483	−0.1818	−0.2945	0.2486
3				0.2654	−0.0985	−0.2200
4					0.2113	−0.0603
5						0.1736

4.3 Integrality: a rank-2 lattice

In the Meixner–Pollaczek case [1] the dual-split coefficients lay in $\mathbb{Z}A \oplus \mathbb{Z}B$ with two Bessel generators. The analogous statement holds here, with generators

$$\mathcal{A} = \sqrt{\pi}\, e^2 E_1(2) = \sqrt{\pi}\, U(1, 1, 2), \qquad \mathcal{B} = \sqrt{\pi}.$$

Conjecture 4.1. *For all m, k (reading the real part when $m+k$ is even, the imaginary part when $m+k$ is odd),*

$$c_{mk} \in \mathbb{Q}\,\mathcal{A} + \mathbb{Q}\,\mathcal{B}.$$

PSLQ at 50-digit precision confirms this for all $m, k \le 5$, with residuals below 10^{-43}:

Table 3: Dual-split coefficients in the lattice $\mathbb{Q}\mathcal{A} + \mathbb{Q}\mathcal{B}$.

(m,k)	c_{mk}	(m,k)	c_{mk}
$(0,0)$	$4\mathcal{A}$	$(2,2)$	$2436\mathcal{A} - 880\mathcal{B}$
$(0,1)$	$20\mathcal{A} - 8\mathcal{B}$	$(2,3)$	$(25660\mathcal{A} - 9272\mathcal{B})/3$
$(0,2)$	$24\mathcal{B} - 68\mathcal{A}$	$(2,4)$	$(27592\mathcal{B} - 76364\mathcal{A})/3$
$(0,3)$	$(208\mathcal{B} - 572\mathcal{A})/3$	$(2,5)$	$(121952\mathcal{B} - 337508\mathcal{A})/5$
$(0,4)$	$(1420\mathcal{A} - 512\mathcal{B})/3$	$(3,3)$	$(106112\mathcal{B} - 293668\mathcal{A})/9$
$(0,5)$	$(16172\mathcal{A} - 5848\mathcal{B})/15$	$(3,4)$	$(338800\mathcal{B} - 937652\mathcal{A})/9$
$(1,1)$	$48\mathcal{B} - 132\mathcal{A}$	$(3,5)$	$(13214836\mathcal{A} - 4774904\mathcal{B})/45$
$(1,2)$	$192\mathcal{B} - 532\mathcal{A}$	$(4,4)$	$353540\mathcal{A} - 127744\mathcal{B}$
$(1,3)$	$(5068\mathcal{A} - 1832\mathcal{B})/3$	$(4,5)$	$(9462964\mathcal{A} - 3419240\mathcal{B})/9$
$(1,4)$	$4628\mathcal{A} - 1672\mathcal{B}$	$(5,5)$	$(88888528\mathcal{B} - 246004652\mathcal{A})/75$
$(1,5)$	$(12400\mathcal{B} - 34316\mathcal{A})/3$		

The denominators observed $(3, 5, 9, 15, 45, 75)$ are odd; a rescaling clearing them, as in the MP case, is expected but not yet identified.

The square root $\sqrt{\zeta'}$ in the generating function (see §6) is not an obstruction but the second generator: since $U(1,1,z) = e^z E_1(z)$ satisfies the inhomogeneous first-order equation $y' = y - 1/z$, the derivative algebra of the full kernel $\sqrt{z}\,U(1,1,z)$ is spanned over $\mathbb{Q}(z)$ by $\{\sqrt{z}\,U(1,1,z),\ \sqrt{z}\}$ — rank two, with the second basis element *algebraic*. This is why $\mathcal{B} = \sqrt{\pi}$ is elementary while $\mathcal{A}$ is transcendental.

(An earlier version of this paper reported that PSLQ finds no relations; the failure was an artifact of the default coefficient bound 10^3, while the integers above reach 10^8.)

5 Decay

5.1 Matrix (a)

The entries b_{0k} for even k decay slowly:

$$|b_{0,0}| = 3.389, \quad |b_{0,2}| = 0.822, \quad |b_{0,4}| = 0.422, \quad |b_{0,6}| = 0.275, \quad |b_{0,8}| = 0.201.$$

The ratios $|b_{0,2(n+1)}|/|b_{0,2n}|$ converge to approximately 0.73, consistent with **power-law decay** $|b_{0,k}| \sim k^{-\alpha}$ with $\alpha \approx 0.7$. The branch point of $\operatorname{sech}(2\pi x)^{3/2}$ at $x = i/4$ is at distance $1/4$ from the real axis, but the polynomial growth of $\widehat{p}_k$ prevents geometric decay.

5.2 Matrix (b)

The entries $|c_{0,k}|$ alternate in magnitude (real vs. imaginary parts) and decay slowly:

$$|c_{0,0}| = 2.562,\ |c_{0,1}| = 1.371,\ |c_{0,2}| = 1.011,\ \ldots,\ |c_{0,9}| = 0.337.$$

The successive ratios converge toward 0.90–0.91, suggesting **power-law decay** $|c_{0,k}| \sim k^{-\beta}$ with $\beta \approx 0.4$. This is significantly slower than geometric decay and contrasts with the MP $\lambda = \frac{1}{4}$ case where the ratios also converge to a constant below 1 but were faster.

6 Generating Function

The Meixner–Pollaczek generating function $\sum_n P_n^{(1/2)}(y;\frac{\pi}{2})\,w^n = (1+w^2)^{-1/2}\,\varphi(w)^{-iy}$, where $\varphi(w) = (1+iw)/(1-iw)$ is the Cayley transform and $P_n^{(1/2)}(2x)$ is related to our $p_n(x)$ by $P_n(2x) = 4^n/(n!)\,p_n(x)$, allows us to sum the double generating function

$$\sum_{m,k\geq 0} I_{mk}\,w^m u^k = \int_{\mathbb{R}} \frac{w_+(x)\,W(x)}{(1+w^2)^{1/2}\,(1+u^2)^{1/2}}\;\varphi(w)^{-2ix}\,\varphi(u)^{-2ix}\,dx,$$

where $I_{mk} = \int P_m(2x)\,P_k(2x)\;w_+\,W\,dx$. After the duplication reduction of §3, this becomes a shifted three-Gamma integral with shift parameter $\zeta' = 2\,\varphi(w)\,\varphi(u)$, yielding

$$\sum_{m,k} I_{mk}\,w^m u^k = \frac{4\pi^{5/2}\,\sqrt{\zeta'/2}\,U(1,1,\zeta')}{\sqrt{(1+w^2)(1+u^2)}} \tag{7}$$

(normalisation verified numerically against the coefficient table to ten digits at $(w,u) = (0.1, 0.15)$). At $w = u = 0$, $\zeta' = 2$, recovering $I_{00} = 4\pi^{5/2}\,U(1,1,2)$ and hence $c_{00} = I_{00}/\pi^2 = 4\sqrt{\pi}\,U(1,1,2)$.

The matching coefficients are $c_{mk} = I_{mk}/\pi^2$ (the conversion factors between P_n and $\widehat{p}_n$ cancel because $\sqrt{h_m h_k} = \pi^2 m!\,k!/4^{m+k}$ exactly absorbs the $4^{m+k}/(m!\,k!)$ from $P_n = (4^n/n!)\,p_n$).

Remark 6.1 (Comparison with the MP case)**.** *The MP generating function [1] had the form $e^{\zeta/2}\,K_0(\zeta/2)/\sqrt{(1-iw)(1-iu)}$, whose kernel generates an irreducible rank-2 module with basis $\{K_0, K_1\}$: both generators A, B are transcendental Bessel values. Here the kernel $\sqrt{z}\,U(1,1,z)$ also generates a rank-2 module, $\{\sqrt{z}\,U(1,1,z), \sqrt{z}\}$, but a reducible one ($e^z E_1(z)$ is an extension of the trivial equation by a Kummer equation), and the reducibility is visible in the lattice: the second generator degenerates to the elementary constant $\mathcal{B} = \sqrt{\pi}$. In both cases the rank of the lattice equals the rank of the derivative module of the full kernel, prefactor included.*

7 Combinatorics

The Euler-number moments $\mu_{2n}/\pi^2 = |E_{2n}|/16^n$ connect the recurrence to the theory of alternating permutations: $|E_{2n}|$ counts the number of alternating permutations of $\{1,\ldots,2n\}$.

The coefficients $\beta_n = n^2/16$ (equivalently $16\beta_n = n^2$, OEIS A000290) identify the weight with the $\lambda = \frac{1}{2}$ MP family.

The first few Euler numbers determining the moments:

$$|E_0| = 1, \quad |E_2| = 1, \quad |E_4| = 5, \quad |E_6| = 61, \quad |E_8| = 1385, \quad |E_{10}| = 50521.$$

8 Zeros

Figure 1 shows the zeros of the truncated dual functions $\sum_{k\leq N} c_{0k}\,\widehat{p}_k(z)$ and $\sum_{k\leq N} b_{0k}\,\widehat{p}_k(z)$ in the complex plane, for $N = 8, 10, 12$.

The $\operatorname{sech}^{3/2}$ zeros are all real, as expected from a positive kernel. The six-Gamma zeros exhibit both real zeros and complex conjugate pairs; as N increases, additional pairs appear, but there is no clear convergence toward a limiting imaginary part (contrast with the MP $\lambda = \frac{1}{4}$ case where zeros converged to $\operatorname{Im} x = -\frac{1}{2}$).

9 Summary

1. The continuous Hahn weight $W = 2\pi^2 \operatorname{sech}(2\pi x)$ has orthogonal polynomials with the exact recurrence $\beta_n = n^2/16$, identified as rescaled Meixner–Pollaczek($\lambda = \frac{1}{2}$, $\phi = \pi/2$).

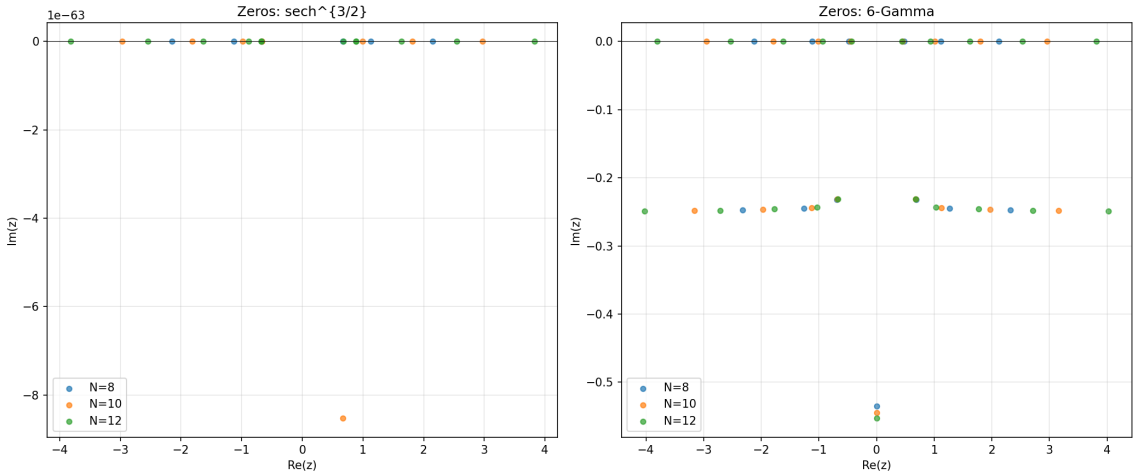


Figure 1: Zeros of the truncated dual functions. Left: $\mathrm{sech}^{3/2}$ kernel (all zeros real). Right: six-Gamma kernel (real and complex zeros).

2. The duplication formula reduces the six-Gamma dual-split integral to a three-Gamma integral, and the shifted three-Gamma lemma gives the **closed form** $c_{00} = 4\sqrt{\pi}\, e^2\, E_1(2)$.

3. The dual-split coefficients lie in the **rank-2 lattice** $\mathbb{Q}\,\mathcal{A} + \mathbb{Q}\,\mathcal{B}$ with $\mathcal{A} = \sqrt{\pi}\, e^2 E_1(2)$, $\mathcal{B} = \sqrt{\pi}$ (verified by PSLQ for $m, k \le 5$). The $\sqrt{\zeta'}$ prefactor supplies the second, algebraic generator: the lattice rank equals the rank of the derivative module of the kernel $\sqrt{z}\,U(1,1,z)$, and the reducibility of that module is why $\mathcal{B}$ is elementary.

4. Both matching matrices exhibit **power-law decay**, not geometric: $|b_{0,k}| \sim k^{-0.7}$, $|c_{0,k}| \sim k^{-0.4}$.

References

[1] R. Langer and Claude Code, *The Dual Meixner–Pollaczek Matching Coefficients: the Calculation*, manuscript, August 2026.